Differentiable Inverse Reachability for RM75: Complete Mathematical Design of the IRDPLAYGROUND Operator

IRDPLAYGROUND technical note

Abstract

This note gives a self-contained mathematical account of the inverse reachability (IRD) operator implemented in `ird_playground`. The operator answers a narrower question than classical discrete IRD tables: given a medically valid tool-centre-point (TCP) pose and a rail-induced robot base (J1-axis) frame, what smooth direction increases collision-checked arm reachability, optionally under a local position/orientation region? We relate the design to five foundational papers (Zacharias et al. 2013; Vahrenkamp et al. 2013; Rudorfer 2024/RM4D; Murooka et al. 2025; Chiu 1988), derive every aggregation formula used in code, and explain why automatic differentiation yields usable IRD gradients, why conditional queries over local pose cones are well-defined, and why the same field supports trajectory-parameter updates without retraining.

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1 Problem statement and classical IRD

1.1 Reachability and inverse reachability

Let the robot configuration be $q \in \mathcal{Q} \subset \mathbb{R}^n$ and let forward kinematics map configurations to TCP poses

$$\mathrm{FK} : \mathcal{Q} \to \mathrm{SE}(3), \qquad q \mapsto T_{\mathrm{tcp}}(q). \tag{1}$$

Here SE(3) is the special Euclidean group of rigid poses (rotation $R \in \mathrm{SO}(3)$ and translation $p \in \mathbb{R}^3$).

Definition 1 (Binary reachability)**.** *A pose $r \in \mathrm{SE}(3)$ is* reachable *if there exists a collision-free configuration q with $\mathrm{FK}(q) = r$ (within IK tolerance) and joint limits satisfied. Write*

$$\mathcal{R} \;=\; \{\, r \in \mathrm{SE}(3) \mid \exists\, q \in \mathcal{Q}_{\mathrm{free}} : \mathrm{FK}(q) = r \,\}. \tag{2}$$

The *capability map* of Zacharias et al. (2013) analyses a robot arm workspace by assigning each workspace voxel a reachability index (and related directional structure) relative to a fixed base. A classical *reachability distribution* (RD) of Vahrenkamp et al. (2013) stores, in a similar voxelisation, a quality score such as the measure of joint configurations that map into that voxel. An *inverse reachability distribution* (IRD) inverts the same data: given a fixed TCP (e.g. a grasp), it yields a distribution over candidate base poses for which that TCP would be reachable.

Formally, if $t_i = T_{\mathrm{tcp},i}^{-1}$ denotes a base-to-TCP sample stored in the RD, then for a world TCP $p \in \mathrm{SE}(3)$ the corresponding candidate base pose is

$$b_i(p) \;=\; p\, t_i^{-1} \;\in\; \mathrm{SE}(3) \tag{3}$$

(with optional projection onto SE(2) for planar base placement). The IRD score of $b_i(p)$ is inherited from the RD quality of t_i. This is an offline discrete table lookup.

1.2 What IRDPLAYGROUND computes instead

Our first-stage operator does *not* replace multi-seed IK or final collision validation. It answers:

> For a medically valid TCP pose T_{tcp} and a rail-induced J1-axis frame T_{axis}, what smooth direction increases collision-checked arm reachability clearance, including under a local position/orientation uncertainty or task region?

Denote the decision variables of interest by x (rail coordinate, TCP yaw on a vessel surface, trajectory control points, ...). We construct a scalar

$$C(x) \;=\; \mathcal{A}\big[\, f\big(c(T_{\mathrm{tcp}}(x), T_{\mathrm{axis}}(x))\big) \,\big] \tag{4}$$

where:

- $c(\cdot)$ is a smooth RM4D-style chart (Section 4);
- f is a learned signed clearance field (Section 5);
- $\mathcal{A}$ is a differentiable set aggregator (Region A, TaskCone, SetQuery, or Trajectory; Sections 7–9).

Gradient-based updates use $\nabla_x C$ from automatic differentiation through the entire Torch graph.

2 Design narrative: from IRD constraints to oriented queries

This section tells the design in *dependency order*: which hard constraints define the IRD here; what a quotient space is; why only J1 may be eliminated; how IRD lives on the rail / J1 axis; why a flange 9-D embedding is selected; how the operator is layered; which tricks keep it differentiable and semantically correct; and finally the *ordered* rotation and lookup steps when tip×roll freedom is present. Later sections supply formulae; this one keeps the reasoning continuous.

2.1 IRD constraints in this problem

Classical IRD (Vahrenkamp et al.) asks: *with the TCP fixed, where should the base sit?* For ultrasound scanning the same question is narrowed by:

1. **Task TCP is semi-fixed.** The contact point comes from anatomy/planning. Allowed freedom is mainly tip tilt and probe roll (controller-choosable), plus *small* pose uncertainty from medical registration (not choosable; must be robustified).

2. **The base is not free in the plane.** The arm rides a linear rail, so base pose is not free in SE(2); only the rail coordinate y remains (plus locked rail geometry).

3. **Queries must be relative to the base.** A fixed world TCP with a sliding rail changes the geometry seen by the arm; reachability must be stated on $T_{\text{axis}\leftarrow\text{tcp}} = (T_{\text{axis}})^{-1}T_{\text{tcp}}$, not on a world-only table.

4. **Clearance includes collision and limits.** The field approximates a signed margin after self-/environment collision and joint-limit checks, not pure kinematic binary reachability.

5. **First-stage guidance, not the final answer.** The operator returns a smooth $C(x)$ and ∇C to move the rail / twist pose; multi-seed IK with hard collision remains decisive.

Under these constraints, "IRD" is no longer a planar base heatmap but a *conditional reachability field along the one-dimensional rail family y (and over task-orientation samples) for a fixed or slowly varying TCP.*

2.2 What a quotient space is

A relative flange pose $g \in \text{SE}(3)$ has six degrees of freedom. If a continuous symmetry group G acts on g and clearance is approximately constant on each orbit, there is no need to store every group element: identify points on the same orbit to obtain the quotient $\text{SE}(3)/G$. Intuitively, a point $[g]$ is "the geometry that remains after removing a known symmetry".

Here $G = \text{Yaw}_{J1}$: simultaneous rotation of position and orientation about the vertical J1 axis (Eq. (15)). Poses on one orbit describe the same arm–task geometry in the axis frame, only spun about gravity. The quotient

$$\mathcal{Q} = \text{SE}(3)/\text{Yaw}_{J1}$$

has intrinsic dimension $6 - 1 = 5$ (Eq. (18)). Training and queries run on $[g]$, which *eliminates* that redundant dimension rather than discarding a physical DoF.

2.3 Why only J1 may be eliminated

Rudorfer's RM4D approximately quotients both *base yaw* and *wrist roll* for many 6-/7-axis arms, yielding a $\sim$4-D map. This system enforces only the first:

- **J1 yaw is a task-audited approximate symmetry.** With a vertical axis and relative axis-frame quantities, a global spin about gravity does not change collision topology or relative joint geometry (under locked rail, ignoring strongly non-revolute environment clutter). Eliminating it compresses samples and lets one field serve forward RM and inverse IRD.

- **Flange / TCP roll is not a symmetry for probe45.** The acoustic axis is offset by $\sim 50°$ from J7. Roll about flange z changes the swept probe volume and self-collision envelope, hence clearance. Quotienting roll as in the legacy 5-D TCP chart would falsely identify distinct task poses.

- **Rail translation is not yaw.** Sliding along e_a changes cylindrical radius/height and lands on a different $[g]$; quotienting J1 does *not* remove the rail DoF.

Thus the quotient is 5-D, not 4-D; roll must remain in the chart so tip×roll queries can tell poses apart.

2.4 How IRD is represented on J1 / the rail

Base placement is parameterised by the axis frame $T^W_{\text{axis}}(y)$ (Eq. (10)): y produces a pure rail translation with fixed orientation factors. For fixed world TCP,

$$y \;\mapsto\; T_{\text{axis}\leftarrow\text{flange}}(y) \;\mapsto\; [g(y)] \in \mathcal{Q} \;\mapsto\; c(g(y)) \in \mathbb{R}^9 \;\mapsto\; f_\theta(c)$$

is the *rail IRD slice* $C_{\text{rail}}(y)$ (Eq. (12)). In other words, IRD is a relative query along the one-parameter family of J1 axis frames, not a discrete table over planar bases; J1 yaw is already removed by the quotient, so base heading need not be scanned. Optimising $\partial C/\partial y$ means "slide the rail toward better clearance".

2.5 Why a flange 9-D embedding is selected

The intrinsic object is $[g] \in \mathcal{Q}$ (5-D), but the network needs a smooth Euclidean input.

1. **Why flange, not TCP.** The chart must split quantities invariant to the last joint from roll-carrying ones; those axes should align with J7 (link_7). The TCP is not coaxial with J7 under the tool offset, so the same split on TCP leaks roll into "invariant" channels.

2. **Why not five minimal coordinates.** Euler / azimuth charts are singular on-axis and branch-cut; AD and boundary stencils break.

3. **Why redundant nine dimensions in production.** Eq. (23) uses cylindrical (p_z, r) with ε, a J7-invariant block, a roll-carrying block, and a ninth component completing $R^\mathsf{T}\hat{z}$, smoothly embedding $\mathcal{Q}$ into $\mathbb{R}^9$. It is constant on yaw orbits (Eq. (25)) and sensitive to roll; extra coordinates are redundancy, not new physics.

The field is $f_\theta : \mathbb{R}^9 \to \mathbb{R}$ (signed clearance). The same field serves "forward: fixed axis, vary TCP" and "inverse: fixed TCP, vary y" because the input is always the relative chart c.

2.6 How the operator is layered

Three layers stack; upper layers never rewrite lower geometry conventions:

Layer	Object	Role
Geometry	T^W_{tcp}, $T^W_{\text{axis}}(y)$, c	Relative pose, rail, TCP→flange, 9-D quotient embedding
Field	$f_\theta(c)$	Pointwise signed clearance (normalise, Fourier, residual MLP, support guard)
Query $\mathcal{A}$	Region A / TaskCone / SetQuery / Trajectory	Aggregate over *sets* of poses (softmin / softmax / mean)

Decision variables x (y, vessel angle, trajectory controls, ...) enter only through geometry into c, then through f and $\mathcal{A}$ into the scalar $C(x)$. The whole graph is differentiable, so $\nabla_x C$ is available for first-stage guidance.

2.7 Key tricks

- **Relative queries:** IRD semantics live on $(T_{\text{axis}})^{-1}T_{\text{tcp}}$; rail motion needs no retraining.
- **Quotient only true symmetries:** J1 yaw only; probe45 roll stays.
- **Redundant smooth chart:** 9-D embedding avoids axis singularities; ε softens the r kink.
- **Signed clearance + boundary stencils:** avoids flat exterior gradients of pure classifiers (Murooka's caveat).
- **Normalisation + Fourier features:** metric and orientation channels share scale; sharp reachable frontiers are representable.
- **Softplus MLP:** C^∞ with nowhere-vanishing derivatives.
- **Support-domain guard:** penalise outside the training box.
- **Split aggregation semantics:** uncertainty $\to$ softmin/CVaR; tip$\times$roll freedom $\to$ softmax (best-of), mean (Eq. (49); cone expectation), or large-τ soft$*$; nested queries reduce uncertainty *before* freedom.
- **Trajectory reduction order:** scenarios $\to$ angles $\to$ path interiors $\to$ full-trajectory softmin (Section 9.1); reversing order flips gradient meaning.

2.8 Ordered tip$\times$roll query pipeline (must not be reordered)

Ultrasound may change tip and roll about a fixed contact point. Local angular increments live in the *medical TCP body* $\{b, t, n\}$: $\delta\omega = (\rho\cos\phi,\ \rho\sin\phi,\ \gamma)$ (Eq. (54)). From a nominal TCP to the scalar C the steps are *fixed* as follows; swapping any step corrupts relative pose or roll semantics.

1. **Nominal world TCP.** Take the medical contact pose $T^{W(0)}_{\text{tcp}}$ (position = contact; orientation = nominal normal/holder).
2. **Rail $\to$ axis frame.** Build $T^{W}_{\text{axis}}(y)$ from the decision y via (10) (translation only; orientation factors fixed).
3. **Sample tip$\times$roll increments (body coordinates).** Draw (ρ, ϕ, γ) in the cone; form $\delta\omega_k \in \mathbb{R}^3$. These are still $\mathfrak{so}(3)$ vectors in the TCP body—not yet multiplied into the world frame.
4. **Right-multiply the exponential (order fixed).**

$$R_k \;=\; R^{(0)}_{\text{tcp}}\,\mathrm{Exp}(\delta\omega_k), \qquad p_k \;=\; p^{(0)}_{\text{tcp}}.$$

   Equivalently $T_k = T^{(0)} \cdot \mathrm{Exp}(\widehat{\delta\omega_k})$: *first* the nominal orientation, *then* tip/roll in the body. Do not use $\mathrm{Exp}(\delta\omega)\,R^{(0)}$ (left multiply = world-frame tilt, different meaning). Position stays at the contact point.

5. **Express relative to the axis.** $T_{\text{axis}\leftarrow\text{tcp},k} = (T^{W}_{\text{axis}}(y))^{-1}T_k$ (Eq. (9)).
6. **TCP $\to$ flange.** Constant mount transform (13) yields $g_k = (p_{\text{fl}}, R_{\text{fl}})_k$.
7. **Quotient embedding.** $c_k = c(g_k) \in \mathbb{R}^9$ (invariant to J1 yaw; sensitive to the tip/roll just applied).
8. **Field evaluation.** $F_k = f_\theta(c_k)$ (or guarded $\tilde{f}$).
9. **Aggregate by query semantics (still ordered).**

- Pure TaskCone: $C = \mathrm{softmax}_\tau\{F_k\}$ (best tip×roll), or $C = C_{\mathrm{mean}}$ (Eq. (56); cone expectation / uniform average).
- If Region A uncertainty wraps freedom: for each free sample η_k, softmin/CVaR over uncertain δ_j first, then softmax / mean over k (Eq. (59): *uncertainty before freedom*).
- Trajectories: softmin over waypoints/path samples next (Section 9.1).

10. **Backpropagation.** $\partial C/\partial y$, $\partial C/\partial \theta$, ... flow through the same ordered graph; swapping left/right multiply or aggregation order aims gradients at the wrong geometric DoF.

In one line: *IRD constraints collapse the base to rail y; the quotient removes only J1 yaw; the 9-D flange chart smoothly represents the remaining 5-D; the field scores points; the query layer handles tip×roll by "right-multiply rotations → relative → flange → c → f → prescribed aggregation order".*

3 Paper map: what each reference contributes

Paper	Core idea absorbed
Zacharias et al. 2013	Capability map: voxel reachability index (+ directional structure)
Vahrenkamp et al. 2013	IRD = fix TCP, score base poses
Rudorfer RM4D 2024	Quotient common yaw/roll symmetries →∼4D map shared by RM & IRD
Murooka et al. 2025	Differentiable reachability map $f_R(r)$ for optimisation; classifiers can have flat exteri
Chiu 1988	Postures should match task transmission requirements

Murooka's differentiable reachability map. Murooka et al. define a scalar $f_R(r)$ with

$$f_R(r) \geq 0 \text{ if reachable}, \qquad f_R(r) < 0 \text{ otherwise}, \tag{5}$$

trained as a smooth binary classifier (e.g. softplus logistic loss). When f_R is continuous and differentiable w.r.t. task coordinates, the gradient $\partial f_R/\partial r$ can enter trajectory optimisation without repeated IK. They also note that a pure decision function may become nearly constant far outside the reachable set, yielding uninformative gradients. Our design follows the same sign convention but supervises an explicit signed clearance with local collision-checked stencils and boundary normals, so that ∇f remains informative near the frontier (Section 6).

RM4D symmetry. Rudorfer observes that for many 6-/7-axis arms, reachability is approximately invariant to base yaw about gravity and to TCP roll about the last joint, reducing an SE(3) query to an intrinsic 4-D quotient. We adopt the dimensionality-reduction spirit but enforce only the symmetries that hold for RM75 with the probe45 tool: base (J1) yaw is a true free symmetry; TCP/flange roll is *not*, because the TCP z-axis is offset by $\sim 50°$ from the J7 axis.

Chiu's task compatibility. Chiu measures how well a posture's velocity/force ellipsoids align with a task. We do not compute Yoshikawa indices online; instead we encode task freedom (large tip×roll cone) and medical uncertainty (tiny box×cone) as set queries over f, so that C is a task-conditioned reachability margin.

4 Frames, rail base, and the flange chart

This section is the geometric backbone of the operator: how a world TCP and a rail coordinate become a yaw-invariant flange feature vector. The chain is

$$(T^W_{\mathrm{tcp}}, y) \xrightarrow{(10)} T^W_{\mathrm{axis}}(y) \xrightarrow{(8)} T_{\mathrm{axis}\leftarrow\mathrm{tcp}} \xrightarrow{(13)} (p_{\mathrm{fl}}, R_{\mathrm{fl}}) \xrightarrow{(23)} c \in \mathbb{R}^9 \xrightarrow{f_\theta} \text{clearance}. \tag{6}$$

Classical IRD scores candidate *bases* for a fixed TCP. Here the base is constrained to a rail, so the free placement variable collapses to the scalar y; the relative transform and the quotient chart below turn that 1-D placement into a query on a single trained field.

4.1 World poses and relative pose

A homogeneous pose is written

$$T \;=\; \begin{bmatrix} R & p \\ 0^\mathsf{T} & 1 \end{bmatrix} \in \mathrm{SE}(3), \qquad R \in \mathrm{SO}(3),\; p \in \mathbb{R}^3. \tag{7}$$

Given world TCP T^W_{tcp} and world J1-axis frame T^W_{axis}, the TCP expressed in the axis frame is

$$\begin{aligned} R_{\mathrm{axis}\leftarrow\mathrm{tcp}} &= (R^W_{\mathrm{axis}})^\mathsf{T}\, R^W_{\mathrm{tcp}}, \\ p_{\mathrm{axis}\leftarrow\mathrm{tcp}} &= (R^W_{\mathrm{axis}})^\mathsf{T}(p^W_{\mathrm{tcp}} - p^W_{\mathrm{axis}}). \end{aligned} \tag{8}$$

Interpretation. Equation (8) is the group product $(T^W_{\mathrm{axis}})^{-1} T^W_{\mathrm{tcp}}$. Equivalently, if both poses are written as elements of SE(3),

$$T_{\mathrm{axis}\leftarrow\mathrm{tcp}} \;=\; (T^W_{\mathrm{axis}})^{-1}\, T^W_{\mathrm{tcp}} \;=\; \begin{bmatrix} R_{\mathrm{axis}\leftarrow\mathrm{tcp}} & p_{\mathrm{axis}\leftarrow\mathrm{tcp}} \\ 0^\mathsf{T} & 1 \end{bmatrix}. \tag{9}$$

All reachability queries are posed in this relative frame, which is exactly the IRD viewpoint: move the base/axis and the relative TCP changes even if the world TCP is fixed. Conversely, a pure world-frame TCP table would recompute reachability whenever the rail slides; the relative product absorbs that motion into a change of $T_{\mathrm{axis}\leftarrow\mathrm{tcp}}$ only.

Why the axis frame, not "base_link" alone. The J1 axis frame is the natural body in which yaw about gravity acts as a left (or simultaneous) rotation of position and orientation. Scoring in that frame makes the subsequent yaw quotient (15) a clean group action rather than a mixture of rail offset and joint-1 angle.

4.2 Rail-induced axis frame

For a linear rail along local axis e_a (code default $a = 1$, i.e. y),

$$T^W_{\mathrm{axis}}(y) \;=\; T_{\mathrm{world}\leftarrow\mathrm{rail}} \underbrace{\begin{bmatrix} I_3 & y\, e_a \\ 0^\mathsf{T} & 1 \end{bmatrix}}_{\text{pure translation by rail coordinate } y} T_{\mathrm{rail}\leftarrow\mathrm{axis}}(y{=}0). \tag{10}$$

Interpretation. y is the raw URDF rail coordinate. Differentiating (10) w.r.t. y moves only the axis origin along the rail; rotation is constant:

$$\frac{\partial p^W_{\mathrm{axis}}}{\partial y} \;=\; R_{\mathrm{world}\leftarrow\mathrm{rail}}\, e_a, \qquad \frac{\partial R^W_{\mathrm{axis}}}{\partial y} \;=\; 0. \tag{11}$$

This is the primitive that turns "IRD w.r.t. free base placement in SE(2) or SE(3)" into "IRD w.r.t. one scalar rail DoF". Substituting $T^W_{\text{axis}}(y)$ into (8) with fixed T^W_{tcp} yields a curve $y \mapsto T_{\text{axis}\leftarrow\text{tcp}}(y)$ in SE(3); composing with the chart (23) and the field f_θ produces the *rail IRD slice*

$$C_{\text{rail}}(y) \;=\; f_\theta\big(c(T_{\text{axis}\leftarrow\text{flange}}(y))\big) \tag{12}$$

(optionally after TCP→flange and set aggregation). Gradients $\partial C_{\text{rail}}/\partial y$ then point toward better rail stations without a discrete IRD table over planar bases.

Fixed factors. $T_{\text{world}\leftarrow\text{rail}}$ places the rail in the room; $T_{\text{rail}\leftarrow\text{axis}}(y{=}0)$ is the locked-rail mount of the arm on the slider. Neither depends on the decision variable y at query time; only the middle translation does.

4.3 TCP to flange

Let $T_{\text{flange}\leftarrow\text{tcp}}$ be the fixed tool transform from URDF (constant for a given mount: probe45, tcp220, horizontal, …). Flange (link_7) pose in the axis frame is

$$(p_{\text{fl}}, R_{\text{fl}}) \;=\; \text{compose}\big((p_{\text{tcp}}, R_{\text{tcp}}),\; T_{\text{flange}\leftarrow\text{tcp}}\big), \tag{13}$$

i.e. $T_{\text{axis}\leftarrow\text{flange}} = T_{\text{axis}\leftarrow\text{tcp}}\,T_{\text{tcp}\leftarrow\text{flange}}$ with $T_{\text{tcp}\leftarrow\text{flange}} = T^{-1}_{\text{flange}\leftarrow\text{tcp}}$ as stored in the kinematic chain. We score flange geometry rather than TCP geometry so that the chart axes align with the J7 rotation axis (needed for the invariant / roll split below). Different mounts change only the constant tool transform; the same chart c and (re)trained field apply once samples are regenerated.

4.4 Quotient by J1 yaw: what is being identified

The previous subsections already give the relative TCP in the axis frame. The quotient and 9-D embedding act on the *flange* (link_7) relative pose, not directly on the TCP: the TCP is mapped through the fixed tool transform first (Eq. (13)); see the end of this subsection and 4.5.

Write the relative flange pose in the axis frame as

$$g \;=\; (p, R) \;=\; T_{\text{axis}\leftarrow\text{flange}} \;\in\; \text{SE}(3), \tag{14}$$

where

- $p \in \mathbb{R}^3$ is the flange origin in the axis frame;
- $R \in \text{SO}(3)$ is the flange orientation, with columns $u_x, u_y, u_z \in \mathbb{R}^3$ (flange body axes expressed in the axis frame);
- $T_{\text{axis}\leftarrow\text{flange}}$ is obtained by composing $T_{\text{axis}\leftarrow\text{tcp}}$ with the fixed $T_{\text{tcp}\leftarrow\text{flange}}$ (at query time the TCP is known, hence g is known).

Simultaneous yaw about the J1/$\hat{z}$ axis acts by

$$\alpha \cdot g \;=\; \big(R_z(\alpha)\,p,\; R_z(\alpha)\,R\big), \qquad R_z(\alpha) \;=\; \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{15}$$

where

- $\alpha \in [0, 2\pi)$ is the rotation angle about the vertical axis (axis-frame $\hat{z}$, aligned with J1)—*not* the rail coordinate y, and *not* an independent network feature;
- $R_z(\alpha)$ is planar rotation about $\hat{z}$;

- $\alpha \cdot g$ is the same relative geometry spun by α about gravity (position and orientation both left-multiplied by R_z).

For RM75 with a vertical J1, this is (approximately) a free symmetry of collision-checked reachability: rotating the whole arm+task about gravity does not create a new kinematic situation once quantities are expressed relative to the axis.

The *yaw orbit* of g is

$$\mathcal{O}(g) \;=\; \{\, \alpha \cdot g \;:\; \alpha \in [0, 2\pi) \,\}, \tag{16}$$

where $\mathcal{O}(g)$ is the set of all poses that differ from g only by absolute heading about $\hat{z}$.

The *quotient space*

$$\mathcal{Q} \;=\; \mathrm{SE}(3)/\mathrm{Yaw}_{J1} \;=\; \{\, [g] \mid g \sim \alpha \cdot g \,\} \tag{17}$$

where

- Yaw_{J1} is the one-dimensional symmetry group generated by $\{\alpha \mapsto R_z(\alpha)\}$;
- $g \sim \alpha \cdot g$ is the equivalence "same yaw orbit";
- $[g] \in \mathcal{Q}$ is the orbit $\mathcal{O}(g)$ as a single point (reachability distinguishes different $[g]$, not different α inside one orbit);
- $\mathcal{Q}$ is the space of all such classes ("relative pose space after removing absolute J1 yaw").

Dimension count: $\dim \mathrm{SE}(3) = 6$, one continuous symmetry, so

$$\dim \mathcal{Q} \;=\; 5, \tag{18}$$

where 5 is the *intrinsic* dimension (translational invariants after cylindrical reduction, plus two independent orientation degrees once yaw is removed; flange roll about J7 remains inside $\mathcal{Q}$).

What we do *not* quotient. RM4D further quotients TCP/tool roll for many stock flanges, reaching an intrinsic 4-D map. For probe45 the acoustic axis is tilted $\sim 50°$ from J7, so roll about the flange z-axis changes the collision and workspace footprint. Those poses lie on *different* orbits under (15) only and must remain distinguishable—hence a 5-D quotient, not 4-D.

Relation to the rail coordinate. The rail map $y \mapsto g(y) = T_{\text{axis}\leftarrow\text{flange}}(y)$ is a curve in SE(3). Pass to the quotient and the curve becomes $y \mapsto [g(y)] \in \mathcal{Q}$. Here y is still the URDF rail coordinate; what is quotiented is α (heading), not y. Distinct y can land on the same orbit only in degenerate cases; in normal use rail motion changes the cylindrical radius/height or relative heading encoded in $[g]$, which is exactly what the IRD-along-rail scalar (12) should see. Translation along e_a is not a Yaw_{J1} action.

4.5 Why the chart uses flange axes u_x, u_y, u_z, not TCP

The query entry point remains the medical/task **TCP**: tip×roll samples, Region A, and trajectory waypoints all operate on T^W_{tcp}. Immediately before the field we apply the fixed step

$$T^{\text{axis}}_{\text{tcp}} \xrightarrow{\;T_{\text{flange}\leftarrow\text{tcp}}\;} g \;=\; T_{\text{axis}\leftarrow\text{flange}}, \tag{19}$$

where $T_{\text{flange}\leftarrow\text{tcp}}$ is mount-dependent (probe45 / tcp220 / horizontal) but constant for a given mount.

The chart vectors u_x, u_y, u_z are columns of the **flange** orientation R_{fl}, because:

- J7 spins about flange z (u_z); splitting a J7-invariant block from a roll-carrying block requires axes aligned with link_7.

- The TCP is not coaxial with J7 under the tool offset; doing the same split on TCP columns leaks roll into "invariant" channels (wrong for probe45).
- The TCP is not discarded: it uniquely determines g via (19); changing the mount only changes the constant tool transform, not the form of c.

4.6 Nine-dimensional yaw-invariant flange embedding

An intrinsic 5-D manifold does not by itself give a network a globally smooth Euclidean input. Charts with Euler angles or a minimal 5-tuple introduce singularities (gimbal lock, branch cuts, undefined azimuth on the J1 axis). We therefore construct a *redundant smooth embedding*

$$c : \mathcal{Q} \longrightarrow \mathbb{R}^9, \tag{20}$$

where c maps each class $[g]$ to a point in $\mathbb{R}^9$, constant on yaw orbits, sensitive to flange roll, and C^∞ (after the mild radial regularisation below) even as $p_{xy} \to 0$.

For a given $g = (p, R)$, write

$$R \;=\; \begin{bmatrix} u_x & u_y & u_z \end{bmatrix}, \qquad p = (p_x, p_y, p_z), \tag{21}$$

where u_x, u_y, u_z are the flange body axes in the *axis frame* (not TCP axes; the TCP was folded into g above). Define the radial smoothed distance

$$r \;=\; \sqrt{p_x^2 + p_y^2 + \varepsilon}, \qquad \varepsilon = (10^{-3}\,\mathrm{m})^2, \tag{22}$$

where r is the cylindrical radius from the J1 axis to the flange origin and ε prevents a derivative blow-up at $r = 0$.

The chart is the map

$$c(p, R) \;=\; \begin{pmatrix} p_z \\ r \\ u_x \cdot \hat{z} \\ p_{xy} \cdot u_{x,xy} \\ p_{xy} \times u_{x,xy} \\ u_z \cdot \hat{z} \\ p_{xy} \cdot u_{z,xy} \\ p_{xy} \times u_{z,xy} \\ u_y \cdot \hat{z} \end{pmatrix} \in \mathbb{R}^9, \tag{23}$$

where

- $p_{xy} = (p_x, p_y)$ is the horizontal position;
- $u_{\cdot,xy}$ is the xy-projection of the corresponding flange axis;
- $\hat{z} = (0, 0, 1)$ is the vertical unit vector in the axis frame (J1 direction);
- the 2-D "cross" $p_{xy} \times u_{xy} := p_x u_y - p_y u_x$ is a scalar;
- the dot/cross terms encode yaw-invariant "heading relative to the horizontal radial direction".

and

$$u_y \cdot \hat{z} \;=\; (u_z \times u_x) \cdot \hat{z} \;=\; u_{z,x}\, u_{x,y} - u_{z,y}\, u_{x,x}, \tag{24}$$

where the last identity uses $u_z \perp u_x$ and the right-handed frame to complete the ninth component and remove a near-axis orientation ambiguity.

Why each component.

- p_z: height along the J1 axis (invariant to yaw about $\hat{z}$).
- r: cylindrical radius; ε removes the kink of $\sqrt{p_x^2 + p_y^2}$ at the axis (curvature $\sim 10^3$/m instead of $\sim 10^6$/m).
- Components involving u_z: J7-invariant block (indices $\{0, 1, 5, 6, 7\}$ in code), because spinning about J7 rotates u_x, u_y but not u_z.
- Components involving u_x and $u_y \cdot \hat{z}$: *roll-carrying* block. They retain flange roll γ, which is a genuine DoF for probe45 and must not be quotiented.
- The ninth entry $u_y \cdot \hat{z}$ completes $R^\mathsf{T} \hat{z}$. Without it, as $r \to 0$ the mixed products vanish and a twofold ambiguity remains in reconstructing orientation about the axis.

Symmetry / embedding statement. If $R_z(\alpha)$ acts as in (15), then

$$c\big(R_z(\alpha)\, p,\ R_z(\alpha)\, R\big) \ = \ c(p, R). \tag{25}$$

Hence c factors through the quotient: $c(g) = c(\alpha \cdot g)$, and we may view c as a map $\mathcal{Q} \to \mathbb{R}^9$. The image is a 5-dimensional submanifold (up to the radial smoothing) sitting in $\mathbb{R}^9$; the four "extra" coordinates are redundancy that buys smoothness and an easy MLP input, not four extra physical degrees of freedom.

Conversion summary. Putting the pieces together for a fixed world TCP and rail station y:

1. Form $T^W_{\text{axis}}(y)$ by (10) (1-D base placement).
2. Relative TCP: $T_{\text{axis}\leftarrow\text{tcp}} = (T^W_{\text{axis}}(y))^{-1} T^W_{\text{tcp}}$ (9).
3. Relative flange via the constant mount transform (13).
4. Quotient feature $c \in \mathbb{R}^9$ (23), invariant to J1 yaw, sensitive to flange roll.
5. Clearance $f_\theta(c)$; optional Region A / TaskCone / trajectory aggregators act on samples of c (or of y and task angles).

Steps 1–2 are the IRD-on-rail reduction; steps 3–4 are the RM4D-inspired quotient embedding specialised to probe45; step 5 is the learned signed field.

Remark 1 (Legacy 5-D TCP chart)**.** *An older embedding* $c_{\text{TCP}} = [p_z,\ u_z\hat{z},\ \|p_{xy}\|,\ p_{xy}u_{xy},\ p_{xy}\times u_{xy}]$ *quotiented both yaw and TCP roll (intrinsic 4-D after both symmetries). For probe45 this is incorrect and is kept only for audits.*

5 Signed reachability field

5.1 Definition

Definition 2 (Signed clearance field)**.** *The network* $f_\theta : \mathbb{R}^9 \to \mathbb{R}$ *approximates a signed reachability clearance on the flange chart:*

$$f_\theta(c) \begin{cases} > 0 & \textit{reachable (positive margin)}, \\ = 0 & \textit{approximate boundary}, \\ < 0 & \textit{unreachable}. \end{cases} \tag{26}$$

This matches Murooka's sign pattern (5) but the *magnitude* is trained to track a collision-checked local distance in a declared metric (Section 6.1), not merely a classification margin.

5.2 Input normalisation and Fourier features

Training chart rows determine centre $\mu \in \mathbb{R}^9$ and scale $\sigma \in \mathbb{R}^9$ (quantile mid-range). Normalised inputs are

$$\tilde{c} \;=\; \frac{c-\mu}{\sigma} \qquad \text{(elementwise).} \tag{27}$$

Fourier features with B bands (B=3 by default) are

$$\Phi(\tilde{c}) \;=\; \Big[\tilde{c},\ \sin(\pi 2^0 \tilde{c}),\ \cos(\pi 2^0 \tilde{c}),\ \ldots,\ \sin(\pi 2^{B-1} \tilde{c}),\ \cos(\pi 2^{B-1} \tilde{c})\Big] \in \mathbb{R}^{9(1+2B)}. \tag{28}$$

Interpretation. Metres and dimensionless orientation channels have incompatible dynamic ranges; without (27), $\sin(\pi 2^k x)$ would scramble them. Fourier lifting helps represent sharp reachable/unreachable transitions without a spatial hash grid (the previous model memorised boundary samples).

5.3 Network architecture

Let $\mathrm{softplus}_\beta(z) = \beta^{-1} \log(1 + e^{\beta z})$. The field is a residual MLP with Softplus activations and a linear scalar head:

$$\begin{aligned} h_0 &= \mathrm{softplus}_\beta(W_0 \Phi(\tilde{c}) + b_0), \\ h_{\ell+1} &= \mathrm{softplus}_\beta(h_\ell + W_\ell^{(2)}\, \mathrm{softplus}_\beta(W_\ell^{(1)} h_\ell + b_\ell^{(1)}) + b_\ell^{(2)}), \\ f_\theta(c) &= w^\mathsf{T} h_L + b. \end{aligned} \tag{29}$$

Softplus (unlike ReLU) is C^∞ with nowhere-vanishing derivative for finite inputs, which is desirable when $\nabla_c f$ is used as a guidance direction.

5.4 Support guard

Let $[\underline{c}, \overline{c}]$ be the axis-aligned box of training chart rows. After normalisation, a smooth exterior distance is

$$d_{\mathrm{supp}}(c) \;=\; \sqrt{\big\|\mathrm{ReLU}(\underline{\tilde{c}} - \tilde{c}) + \mathrm{ReLU}(\tilde{c} - \overline{\tilde{c}})\big\|_2^2 + \epsilon^2} - \epsilon, \tag{30}$$

and the guarded score is

$$\tilde{f}(c) \;=\; f_\theta(c) - \lambda_g\, d_{\mathrm{supp}}(c), \qquad \lambda_g = 8 \text{ (default).} \tag{31}$$

Interpretation. Outside the training support, $\tilde{f}$ is pulled negative so optimisation is discouraged from extrapolating into unlabelled chart regions.

5.5 World scoring (the IRD point query)

$$F(T^W_{\mathrm{tcp}}, T^W_{\mathrm{axis}}) \;=\; \tilde{f}\Big(c\big(\mathrm{flange}((T^W_{\mathrm{axis}})^{-1} T^W_{\mathrm{tcp}})\big)\Big). \tag{32}$$

Interpretation. Fix T^W_{tcp} and vary $T^W_{\mathrm{axis}}(y)$ via (10): then F is precisely a smooth IRD score of the rail/base placement. Fix the axis and vary T^W_{tcp}: then F is a smooth forward reachability score. One field serves both views because of the relative chart (23).

6 Declared SE(3) metric and training targets

6.1 Metric

Boundary labels use the weighted distance that equates 1 cm of translation with 1° of rotation:

$$\lambda \;=\; \frac{0.01\,\mathrm{m}}{1^\circ \text{ in radians}} \;=\; \frac{0.01}{\pi/180} \;\approx\; 0.5730\,\mathrm{m/rad}. \tag{33}$$

For a translational increment $\Delta p \in \mathbb{R}^3$ (metres) and rotational increment $\Delta\omega \in \mathbb{R}^3$ (radians, rotation-vector / so(3) coordinates),

$$d(\Delta p, \Delta\omega) \;=\; \sqrt{\|\Delta p\|_2^2 + \lambda^2 \,\|\Delta\omega\|_2^2}. \tag{34}$$

Interpretation. This is *not* the bi-invariant SE(3) geodesic; it is a declared chart-friendly metric used to place stencil samples and to interpret clearance in "metres".

6.2 Stencil supervision

Around collision-checked boundary centres one evaluates exact reachable / unreachable labels at offsets

$$\pm 1, 3, 6, 10\,\text{mm} \quad \text{and} \quad \pm\, 1, 3, 5^\circ, \tag{35}$$

plus explicit zero-valued bisected boundary centres. These provide targets $s^\star(c)$ for the signed value of $f_\theta(c)$.

6.3 Losses

Let $\{(c_i, y_i, s_i^\star, n_i)\}_i$ be a minibatch with binary label $y_i \in \{0, 1\}$, SDF target $s_i^\star$, and (when available) unit boundary normal n_i in normalised chart coordinates.

Classification (sign).

$$\mathcal{L}_{\text{cls}} \;=\; \frac{\sum_i w_i^{\text{cls}}\, \ell_{\text{BCE}}(\alpha f_\theta(c_i),\, y_i)}{\sum_i w_i^{\text{cls}}}, \tag{36}$$

where α is a logit scale (default 3) and ℓ_{BCE} is binary cross-entropy with logits. *Interpretation.* Enforces Murooka-style sign correctness (26).

Signed value (magnitude). On rows with SDF weight $w_i^{\text{sdf}} > 0$,

$$\mathcal{L}_{\text{sdf}} \;=\; \text{SmoothL1}_{\beta=0.1}(f_\theta(c_i),\, s_i^\star). \tag{37}$$

Interpretation. Makes $|f|$ meaningful as clearance, not an arbitrary classifier margin.

Boundary normal alignment. With $\tilde{c}$ requiring grad,

$$\mathcal{L}_{\text{n}} \;=\; \mathbb{E}_i\Big[1 - \frac{\nabla_{\tilde{c}} f_\theta(c_i) \;\cdot\; n_i}{\|\nabla_{\tilde{c}} f_\theta(c_i)\| \, \|n_i\|}\Big]. \tag{38}$$

Interpretation. Cosine loss aligns the field gradient with the collision-checked crossing direction. This is why ∇F points toward increasing reachability near the frontier rather than along arbitrary level sets.

Empirical boundary slope (production loss).

$$\mathcal{L}_{\text{boundary-slope}} = \mathbb{E}_{(c^-, c^+, \Delta c)}\left[\left(\frac{f(c^+) - f(c^-)}{\|c^+ - c^-\|_2 + \epsilon} - \widehat{s}_{\text{stencil}}\right)^2\right]. \tag{39}$$

This preserves measured stencil direction and slope as independent supervision; it is not an Eikonal prior.

Optional Eikonal (disabled in production).

$$\mathcal{L}_{\text{Eik}} \;=\; \text{SmoothL1}(\|\nabla_{\tilde{c}} f\|,\; v^{\star}), \tag{40}$$

with $v^{\star} = 1$ or an empirical stencil slope. Workspace-normalised mixed units make a universal unit-speed prior poorly conditioned; production relies on (37)–(38) instead.

Total.

$$\mathcal{L} \;=\; \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{sdf}}\mathcal{L}_{\text{sdf}} + \lambda_{\text{n}}\mathcal{L}_{\text{n}} + \lambda_{\text{boundary-slope}}\mathcal{L}_{\text{boundary-slope}} + \lambda_{\text{Eik}}\mathcal{L}_{\text{Eik}} + \cdots \tag{41}$$

Remark 2 (Why not a classifier alone?)**.** *A classifier can achieve high accuracy while being nearly constant for c deep in the unreachable region, so $\nabla_x F \approx 0$ exactly when an optimiser most needs a recovery direction (Murooka's caveat). Signed stencil values plus normal alignment keep a usable slope across the boundary band.*

6.4 Held-out accuracy of point queries f_θ

Table 1 summarises the production checks that establish f_θ as a usable optimization-direction operator (`rm4d_signed_production`). Full metrics are in `docs/rm4d_signed_ird_design.md`.

Table 1: Accuracy of f_θ.

Quantity	Value
Balanced accuracy (frozen baseline)	90.38%
Reachable recall	91.94%
Unreachable specificity	88.82%
Positional direction agreement	99.80%
Rotational direction agreement	99.58%
Position strict / wide straddle	23.91%/90.79%
Rotation strict / wide straddle	10.28%/70.44%
Position crossing P95	0.953 mm
Rotation crossing P95	0.0955 deg
Rail AD–FD median relative error	0.0139%
TCP x AD–FD median relative error	0.0074%

Sign agreement supports point queries; direction agreement and AD–FD agreement show that ∇F is a usable update direction. Hard feasibility still requires final IK with collision checking.

7 Differentiable set queries

All set queries share the pattern: sample poses about a nominal TCP, score each with (32), reduce with a smooth aggregator.

7.1 Local pose perturbation

In the medical TCP frame with columns $[b, t, n]$ (binormal, tangent, normal), a position offset $\delta p_{\text{loc}} \in \mathbb{R}^3$ and rotation vector $\delta\omega_{\text{loc}} \in \mathbb{R}^3$ produce

$$\begin{aligned} p' &= p + R\,\delta p_{\text{loc}}, \\ R' &= R\;\text{Exp}(\delta\omega_{\text{loc}}), \end{aligned} \tag{42}$$

where $\text{Exp} : \mathbb{R}^3 \to \text{SO}(3)$ is the Rodrigues map

$$\text{Exp}(\omega) \;=\; I + \frac{\sin\theta}{\theta}[\hat{\omega}] + \frac{1-\cos\theta}{\theta^2}[\hat{\omega}]^2, \qquad \theta = \|\omega\|, \tag{43}$$

and $[\hat{\omega}]$ is the skew-symmetric matrix of the unit axis (with the usual $\theta \to 0$ limit $I + [\omega]$).

7.2 Normalised softmin and softmax

For a vector $v \in \mathbb{R}^K$ and temperature $\tau > 0$,

$$\operatorname{LogSumExp}(v) \;=\; \log \sum_{k=1}^{K} e^{v_k}. \tag{44}$$

The *normalised softmin* used for robust aggregation is

$$\operatorname{softmin}_\tau(v) \;=\; -\tau\Big(\operatorname{LogSumExp}(-v/\tau) - \log K\Big). \tag{45}$$

Why the $\log K$ *term?* If $v_k \equiv a$ for all k, then $\operatorname{LogSumExp}(-v/\tau) = \log K - a/\tau$, hence $\operatorname{softmin}_\tau(v) = a$. A constant field is unchanged; without $\log K$, softmin would shift by $-\tau \log K$.

The *normalised softmax-like maximum* is

$$\operatorname{softmax}_\tau(v) \;=\; \tau\Big(\operatorname{LogSumExp}(v/\tau) - \log K\Big). \tag{46}$$

Same constant-preserving property for maxima.

Implicit weights (gradient structure). Differentiating (45) gives

$$\frac{\partial \operatorname{softmin}_\tau(v)}{\partial v_k} \;=\; \frac{e^{-v_k/\tau}}{\sum_j e^{-v_j/\tau}} \;=\; \pi_k^{\min}(v), \tag{47}$$

i.e. a Gibbs distribution concentrated on the *smallest* entries. For (46),

$$\frac{\partial \operatorname{softmax}_\tau(v)}{\partial v_k} \;=\; \frac{e^{v_k/\tau}}{\sum_j e^{v_j/\tau}} \;=\; \pi_k^{\max}(v), \tag{48}$$

concentrated on the *largest* entries. Thus robust queries backpropagate into worst-case samples; task-free softmax queries backpropagate into best-case samples.

Mean (cone expectation). When angles are treated as a distribution $\mu(\mathrm{d}\eta)$ on the tip×roll cone (uniform in the sampling measure, or Sobol quadrature), the natural soft aggregate is the expectation rather than a soft extremum:

$$C_{\text{mean}} \;=\; \operatorname{mean}(v) \;=\; \frac{1}{K}\sum_{k=1}^{K} v_k \;\approx\; \mathbb{E}_{\eta\sim\mu}\big[F(T(\eta),\, T_{\text{axis}})\big]. \tag{49}$$

Its gradient weights are uniform,

$$\frac{\partial C_{\text{mean}}}{\partial v_k} \;=\; \frac{1}{K}, \qquad \nabla_x C_{\text{mean}} \;=\; \frac{1}{K}\sum_{k=1}^{K} \nabla_x F\big(T(\eta_k),\, T_{\text{axis}}(x)\big). \tag{50}$$

Relation to softmin/softmax. As $\tau \to \infty$, both (45) and (46) flatten toward the sample mean (up to the constant-preserving normalisation). Mean is therefore the honest "average over uncertain / free angles" limit: every sample contributes equally to the update direction, unlike softmin (mass on the worst) or softmax (mass on the best).

7.3 Region A: uncertainty box $\times$ small cone (softmin)

Definition 3 (Region A)**.** *Default extents: binormal* 4 mm*, tangent* 3 mm*, normal* 2 mm*; orientation cone half-angle* $\beta = 3°$*;* $K = 64$ *fixed scrambled Sobol samples including the nominal pose.*

Position half-samples use a joint 5-D Sobol sequence $u \in (0,1)^5$:

$$\delta p \;=\; (2u_{1:3} - 1) \odot (b,t,n)_{\text{extent}}, \tag{51}$$

and orientation samples are uniform in solid angle inside the cone,

$$\begin{aligned} \cos\rho &= 1 - u_4(1-\cos\beta), \\ \rho &= \arccos(\cos\rho), \\ \phi &= 2\pi u_5, \\ \delta\omega &= (\rho\cos\phi,\ \rho\sin\phi,\ 0), \end{aligned} \tag{52}$$

then antisymmetrised $\{\,0,\ \pm\text{half samples}\,\}$ to length K. The Region A clearance is

$$C_{\text{A}}(T_{\text{tcp}}, T_{\text{axis}}) \;=\; \text{softmin}_\tau \Big\{ F\big(T_{\text{tcp}}(\delta_k),\, T_{\text{axis}}\big) \Big\}_{k=1}^{K}, \qquad \tau = 0.15. \tag{53}$$

Interpretation. C_{A} is a smooth lower envelope over a medically motivated uncertainty set U. Increasing C_{A} improves the worst likely pose under small registration / contact error. Coverage diagnostics use $\frac{1}{K}\sum_k \sigma(F_k/\tau)$.

7.4 Task cone: tip $\times$ roll free set (softmax or mean)

Ultrasound contact often permits a large tip tilt and probe roll as *controller-chosen* free variables, not as uncertainty. The same sample set also supports a mean aggregator when tip$\times$roll is treated as *uncertain* (or when the placement should improve the whole cone, not only its best pose).

Definition 4 (TaskCone)**.** *Tip half-angle* β_{tip} *(e.g.* $20°$*–*$45°$*), roll half-range* $\gamma_{\max}$ *(e.g.* $20°$*–*$30°$*),* K *Sobol samples in* (ρ, ϕ, γ)*.*

$$\begin{aligned} \cos\rho &= 1 - u_1(1-\cos\beta_{\text{tip}}), \\ \phi &= 2\pi u_2, \\ \gamma &= (2u_3 - 1)\gamma_{\max}, \\ \delta\omega &= (\rho\cos\phi,\ \rho\sin\phi,\ \gamma). \end{aligned} \tag{54}$$

Position is held fixed (contact point prescribed). Write $v_k = F(T_{\text{tcp}}(\eta_k), T_{\text{axis}})$. Two common aggregates are

$$C_{\text{task}}(T_{\text{tcp}}, T_{\text{axis}}) \;=\; \text{softmax}_\tau\{v_k\}_{k=1}^{K}, \qquad \tau = 0.20, \tag{55}$$

and the cone expectation (49),

$$C_{\text{mean}}(T_{\text{tcp}}, T_{\text{axis}}) \;=\; \mathbb{E}_{\eta\sim\mu}\big[F(T_{\text{tcp}}(\eta),\, T_{\text{axis}})\big] \;\approx\; \frac{1}{K}\sum_{k=1}^{K} v_k, \tag{56}$$

with update direction (50). *Interpretation.* Opposite of Region A: softmax C_{task} asks whether *some* tip$\times$roll choice makes the arm comfortable. Mean C_{mean} asks how good the cone is *on average* and backpropagates equally into every sample. Choose softmax when tip$\times$roll is controller freedom; choose mean (or softmin/CVaR) when angles are truly uncertain or the placement should raise the whole cone. Both express the Chiu-style "match the posture set to the task" idea as a reachability margin rather than an ellipsoid index.

7.5 Nested free × uncertain query (SetQuery)

The general Phase-4 semantic is

$$C_{\text{set}}(T) \;=\; \max_{\eta \in D_{\text{free}}} \; \underbrace{\text{CVaR}_\alpha \text{ or softmin}}_{\delta \in U} \; F\big(c(T(\eta,\delta))\big). \tag{57}$$

Batch scores have shape $[N, K_{\text{free}}, K_{\text{unc}}]$. Reduce the uncertain axis first, then take a smooth max over free variables.

Rockafellar–Uryasev lower-tail CVaR. For clearance values (higher better), lower-tail CVaR at level α is

$$\text{CVaR}_\alpha(X) \;=\; \max_{t\in\mathbb{R}} \left\{ t - \frac{1}{\alpha}\,\mathbb{E}\big[(t-X)_+\big] \right\}. \tag{58}$$

Interpretation. t plays the role of a VaR threshold; the expectation penalises shortfalls below t. Empirically the optimum lies in the sample set, so code evaluates the objective at every sample value and takes a hard or soft maximum over those candidate t's, consistent with the auxiliary maximisation. This is more statistically stable than a pure softmin when K_{unc} is moderate.

Implemented reduction.

$$\begin{aligned} g(\eta) &= \text{CVaR}_\alpha\{F(T(\eta,\delta_j))\}_j \quad \text{or} \quad \text{softmin}_\tau\{\dots\}, \\ C_{\text{set}} &= \text{softmax}_\tau\{g(\eta_i)\}_i. \end{aligned} \tag{59}$$

8 Why IRD gradients exist and are useful

8.1 Chain rule through the IRD point query

Consider rail coordinate y with fixed world TCP. Then

$$\frac{\mathrm{d}F}{\mathrm{d}y} \;=\; \frac{\partial \tilde f}{\partial c}\,\frac{\partial c}{\partial(p,R)}\,\frac{\partial(p,R)}{\partial T_{\text{axis}}}\,\frac{\mathrm{d}T_{\text{axis}}}{\mathrm{d}y}. \tag{60}$$

Every factor is implemented with Torch ops (Exp, matrix products, $\sqrt{\cdot+\varepsilon}$, Softplus MLP), so $\mathrm{d}F/\mathrm{d}y$ is exact up to floating-point error. Empirical AD-vs-FD median relative error on rail and TCP-x is on the order of 10^{-4}.

Geometric reading. $\partial c/\partial(p,R)$ converts a spatial twist of the relative flange pose into chart coordinates. $\partial\tilde f/\partial c$ is trained to align with the outward reachable normal near the boundary (Eq. (38)). Hence $\mathrm{d}F/\mathrm{d}y > 0$ means "moving the carriage in the $+y$ direction increases predicted clearance".

8.2 Gradients of conditional cone queries

For Region A,

$$\nabla_x C_{\text{A}} \;=\; \sum_{k=1}^{K} \pi_k^{\min}\,\nabla_x F\big(T_{\text{tcp}}(\delta_k;x),\,T_{\text{axis}}(x)\big). \tag{61}$$

For TaskCone (softmax),

$$\nabla_x C_{\text{task}} \;=\; \sum_{k=1}^{K} \pi_k^{\max}\,\nabla_x F\big(T_{\text{tcp}}(\eta_k;x),\,T_{\text{axis}}(x)\big). \tag{62}$$

For the cone mean (56),

$$\nabla_x C_{\text{mean}} \;=\; \frac{1}{K}\sum_{k=1}^{K} \nabla_x F\big(T_{\text{tcp}}(\eta_k; x),\, T_{\text{axis}}(x)\big). \tag{63}$$

Interpretation of "conditional IRD gradient". The condition is the set U or D_{free}. The scalar C is the set-aggregated clearance. Its gradient w.r.t. base/rail/path parameters is a *weighted average of point-IRD gradients* at the sampled local poses, with weights (47), (48), or the uniform $1/K$ of (50). No discrete voxel argmax is required.

8.3 From point clearances to conditional queries and update directions

This subsection chains 7.4–8.2 into one intuitive pipeline. *Training* fits only $f_\theta(c)$ (pointwise); *inference* uses $\mathcal{A}$ for region/angle queries, then automatic differentiation for the update direction.

Step 1 — Point query. With a nominal TCP and rail coordinate y, a single pose T scores as

$$v \;=\; F(T, T_{\text{axis}}(y)) \;=\; f_\theta\big(c(T, T_{\text{axis}}(y))\big) \;\in\; \mathbb{R}. \tag{64}$$

Here v is signed clearance at that pose: *no* angle set and *no* direction vector yet.

Step 2 — Region/angle = many points + aggregation. For TaskCone (free tip×roll), draw K increments η_k in the cone, form poses $T_k = T_{\text{tcp}}(\eta_k)$, and evaluate pointwise:

$$\begin{aligned} v_k &= F\big(T_k,\, T_{\text{axis}}(y)\big), \qquad k = 1, \dots, K, \\ C_{\text{task}}(y) &= \operatorname{softmax}_\tau(v_1, \dots, v_K) \;\approx\; \text{how good the best pose in the cone is}, \\ C_{\text{mean}}(y) &= \tfrac{1}{K}\sum_k v_k \;\approx\; \text{how good the cone is on average.} \end{aligned} \tag{65}$$

Region A is the same with softmin (worst-case). Thus

$$\text{region/angle query} \;=\; \text{many pointwise } f \text{ queries} + \text{softmin}\,/\,\text{softmax}\,/\text{mean} \quad \text{(no separate region network)}.$$

Step 3 — Update direction = gradient of the scalar C. Let the decision variable be rail y (or vessel angle θ). The computational graph is

$$y \;\to\; T_{\text{axis}}(y) \;\to\; \{T_k\} \;\to\; \{c_k\} \;\to\; \{v_k = f_\theta(c_k)\} \;\to\; C = \mathcal{A}(v). \tag{66}$$

Automatic differentiation yields

$$\frac{\partial C}{\partial y} \;=\; \sum_{k=1}^{K} \frac{\partial C}{\partial v_k}\,\frac{\partial v_k}{\partial y}, \tag{67}$$

where $\partial C/\partial v_k$ is the softmin/softmax weight π_k from (47) or (48), or the uniform $1/K$ of (50):

- softmin: mass on the *worst* sample ⇒ improve the worst;
- softmax: mass on the *best* sample ⇒ raise an already good pose or move the placement toward a good region;
- mean: equal weight on every sample ⇒ raise the cone average (Eq. (63)).

Quiver arrows / update steps are $\partial C/\partial y$ (or $\partial C/\partial\theta$); they are *not* a collage of per-point "direction features". Each point only has a scalar v_k; the direction is the derivative of C w.r.t. the decision variable (Eqs. (62), (69)).

8.4 Vessel-surface example (θ, y)

On an elliptical vessel, TCP position is tied to surface angle θ and path abscissa s, while rail y sets the base. Demo code forms

$$C(s;\theta,y) \;=\; C_{\text{task}}\big(T_{\text{tcp}}(\theta,s),\, T_{\text{axis}}(y)\big) \tag{68}$$

and extracts

$$\frac{\partial C}{\partial \theta}, \qquad \frac{\partial C}{\partial y} \tag{69}$$

by `torch.autograd.grad`. Quiver plots of these components are literal visualisations of the conditional IRD vector field on the (θ, y) planning plane.

9 Trajectory operator and online updates

9.1 Aggregation order

Given waypoints $\{T_{\text{tcp}}^{(\ell)}\}_{\ell=1}^{W}$ and a (shared or per-waypoint) axis frame, the trajectory operator applies, in order:

1. **Uncertainty / Region A scenarios** for each approximate-angle candidate;
2. **Angle aggregation** (softmax select, softmin robust, mean, or CVaR);
3. **Path residuals**: interior SE(3) samples on each segment;
4. **Trajectory reduction**: softmin / CVaR / hard min over waypoint and path margins.

9.2 SE(3) path interpolation

On a segment from T_0 to T_1, for $\alpha \in (0,1)$,

$$\begin{aligned} p(\alpha) &= (1-\alpha)p_0 + \alpha p_1, \\ R(\alpha) &= R_0\ \mathrm{Exp}(\alpha\ \mathrm{Log}(R_0^{\top} R_1)). \end{aligned} \tag{70}$$

Interpretation. Translations move linearly; orientations follow the geodesic in SO(3). Controllers that require the whole interpolated path to be IK-feasible need these interior samples: endpoint reachability alone is insufficient.

9.3 Per-waypoint and path margins

Let A index angle candidates and S index Region A scenarios. After scoring,

$$\begin{aligned} g_{\ell,a} &= \mathcal{A}_S\big[F(T_{a,s}^{(\ell)})\big], \\ C_\ell &= \mathcal{A}_A\big[g_{\ell,a}\big], \end{aligned} \tag{71}$$

and for interior path samples C_m^{path} similarly. The scalar trajectory clearance is

$$C_{\text{traj}} \;=\; \mathrm{softmin}_\tau\Big[\{C_\ell\}_{\ell=1}^{W} \;\cup\; \{C_m^{\text{path}}\}_m\Big] \tag{72}$$

(or CVaR / hard min). Optional SRS arm angle ψ enters as an outer softmax over ψ-conditioned fields when exposed.

9.4 Why the landscape updates with the trajectory

The network f_θ is trained once in the flange chart and then frozen. A trajectory is only a curve $x \mapsto (T_{\mathrm{tcp}}(x), T_{\mathrm{axis}}(x))$. Because:

1. c depends on the *relative* pose (8);
2. cone samples are expressed in the *current* TCP body frame (42);
3. aggregators (45)–(49) are memoryless functions of the current sample scores;

re-evaluating $C_{\mathrm{task}}(s)$ or $C_{\mathrm{traj}}(x)$ at a new path automatically yields a new IRD landscape and new gradients $\nabla_x C$ *without* modifying θ.

9.5 Typical optimisation objective

Ellipse-vessel demos minimise a hinge toward a conformal clearance target $m^\star$ plus regularisers:

$$\begin{gathered}\ell_{\mathrm{IRD}} = \frac{1}{W}\sum_{\ell} \operatorname{softplus}\Big(\frac{m^\star - C_\ell}{\sigma}\Big) + \max_{\ell} \operatorname{softplus}\Big(\frac{m^\star - C_\ell}{\sigma}\Big) - \kappa\,\overline{C}, \\ \ell = w_{\mathrm{IRD}}\,\ell_{\mathrm{IRD}} + w_{\mathrm{near}}\,\ell_{\mathrm{nearest}} + w_{\mathrm{cont}}\,\ell_{\mathrm{continuity}} + \cdots\end{gathered} \tag{73}$$

Interpretation. $\operatorname{softplus}((m^\star - C)/\sigma)$ is a smooth one-sided penalty: inactive when $C \geq m^\star$, rising when clearance falls below target. The extra $-\kappa\overline{C}$ term continues to reward climbing toward IRD peaks even after the hinge is satisfied. Continuity penalties keep $\theta(s)$ and $y(s)$ smooth. Gradient descent on (73) is exactly "update the trajectory using conditional IRD gradients".

9.6 Fail-closed 8-DOF trajectory SQP

The production planner lifts IRD warm starts with one robot variable per waypoint, $q_i = [y_i, q_{1i}, \dots, q_{7i}] \in \mathbb{R}^8$, and no duplicate rail variable. A six-dimensional offset z_i gives the soft nominal target $T_i(z_i) = \exp(\widehat{z}_i)T_i^{\mathrm{nom}}$. The objective combines pose error, normalised-s first/second differences, limit margin, singularity and early IRD-margin terms. Hard constraints are $\mathrm{FK}(q_i) = T_i(z_i)$, medical position/rotation bands, true rail/joint limits, collision distance, world patient/bed/contact constraints and adaptive interior segment samples. Pinocchio Jacobians, ProxSuite subproblems and HPP-FCL witness-point gradients are finite-difference audited. Only a path with every hard validator passing, FK error at most $0.5\,\mathrm{mm}/1^\circ$, and scaled KKT residual at most 10^{-3} is returned; otherwise the API returns `valid=false` with no substitute path. TCP arc length seeds time at no more than $0.02\,\mathrm{m/s}$; Ruckig may only slow it. The result supplies pose, Cartesian feedforward, rail/joint references, joint feedforward and contact normals to the existing QP-IK/force-position controller. IRD remains guidance, not a feasibility certificate.

10 End-to-end commutative diagram

$$\begin{gathered} x \xrightarrow{\text{kinematics / rail}} (T_{\mathrm{tcp}}(x), T_{\mathrm{axis}}(x)) \xrightarrow{\text{local samples}} \{T_k(x)\} \\ \xrightarrow{(8)+(23)} \{c_k\} \xrightarrow{\tilde{f}} \{F_k\} \xrightarrow{\mathcal{A}} C(x) \xrightarrow{\nabla} \nabla_x C \end{gathered}$$

Proposition 1 (Single autograd tape)**.** *If every arrow above is implemented with elementary differentiable maps, then $\nabla_x C$ exists almost everywhere and coincides with the analytic chain rule. Discrete IRD voxel lookups break the tape at the table index; the neural field does not.*

11 Scope, guarantees, and non-goals

- **Guidance, not certificate.** Held-out boundary errors are sub-millimetre when bracketed, but the field is an optimisation-direction operator. Hard feasibility still requires final multi-seed IK and collision checking.
- **Symmetry approximation.** J1 yaw is a task-audited approximation because of the finite $\pm 177.96°$ limit; the unused seam lies outside this scan task, and final SQP always enforces the true limit.
- **No branch / continuous ψ in the point field.** Offline labels are point reachability. Path residuals and continuous arm-angle selection live in the trajectory operator.
- **The field is not the planner.** Region A / TaskCone / TrajectoryTask provide query layers; the separate `ird_playground.optimization` package performs the fail-closed rail+7R joint lift and hard validation.

12 Symbol table

Symbol	Meaning
$T_{\mathrm{tcp}}, T_{\mathrm{axis}}$	World TCP pose; world J1-axis pose
$c \in \mathbb{R}^9$	Yaw-invariant flange chart
$f_\theta, \tilde{f}$	Signed field; support-guarded field
F	World point score (32)
$C_{\mathrm{A}}, C_{\mathrm{task}}, C_{\mathrm{mean}}, C_{\mathrm{set}}, C_{\mathrm{traj}}$	Aggregated clearances
y	Rail coordinate
δ, η	Uncertainty sample; free (task) sample
μ	Distribution on tip×roll cone (49)
τ	Softmin/softmax temperature
λ	Translation–rotation metric scale (33)
$\pi^{\min}, \pi^{\max}$	Softmin / softmax weights

13 References

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2. N. Vahrenkamp, T. Asfour, and R. Dillmann, “Robot placement based on reachability inversion,” ICRA 2013.
3. M. Rudorfer, “RM4D: A Combined Reachability and Inverse Reachability Map for Common 6-/7-axis Robot Arms by Dimensionality Reduction to 4D,” ICRA 2025 / arXiv:2410.06968.
4. M. Murooka et al., “Learning Differentiable Reachability Maps for Optimization-based Humanoid Motion Generation,” Humanoids 2025 / arXiv:2508.11275.
5. S. L. Chiu, “Task Compatibility of Manipulator Postures,” International Journal of Robotics Research, 1988.
6. Internal design note: `ird_playground/docs/rm4d_signed_ird_design.md`.

A Code $\leftrightarrow$ equation map

Equation	Primary module
(10)	`region/operator.py::base_from_rail_torch`
(23)	`ird/canonical.py::canonical_flange_invariants_torch`
(32)	`neural/signed_field.py::score_world`
(45)–(50)	`region/operator.py`, `region/set_query.py`, `region/trajectory_operator.py`
(53)	`region/operator.py::RegionA`
(55)–(56)	`region/task_cone.py::TaskConeReachability` (`mean_clearance`; mean as opt. aggregate)
(57)–(59)	`region/set_query.py::SetQueryOperator`
(70)–(72)	`region/trajectory_operator.py`
(36)–(41)	`neural/train_signed.py`
(73)	`experiments/ellipse_vessel_ird_demo.py`